RescueNet: Offline-First Disaster Coordination System

Project Summary — Stages 1 to 8 Complete | Zero-Internet Peer-to-Peer Mesh

Python 3.10+ | SQLite Event Sourcing | Ed25519 + AES-256-GCM | TCP LAN + Streamlit UI

1. What We Built

RescueNet enables field teams across disconnected camps (Harbor, Trauma Clinic, Mountain Shelter) to coordinate shelters, missing persons, scarce rescue boats, and divisible medical supplies with zero internet connectivity. Every station stores an append-only SQLite event log locally, synchronizes missing deltas peer-to-peer over Wi-Fi hotspots or carried data mules, and converges deterministically.

2. Summary of Completed Stages (1–8)

Stage 1 — Immutable Events & SQLite `events/event.py, storage/sqlite_store.py, projector.py`

Content-addressed SHA-256 Events, append-only SQLite store (anti-equivocation + deduplication), and deterministic StateProjector with canonical (timestamp, node_id, counter) ordering.

Stage 2 — Anti-Entropy Delta Sync `sync/node.py, sync/messages.py, transport/in_memory.py`

Bi-directional version-vector sync protocol exchanging only missing delta events, transitive gossip convergence, and exact wire byte accounting.

Stage 3 — Scarce Allocation Safety `allocation/escrow.py, tokens.py, conflict_resolver.py`

Escrow quotas prevent supply over-allocation offline; single-owner token prevents double-booking boats; conflict ledger + human triage prevents silent lost updates.

Stage 4 — N-Node Partition Simulator `simulator/engine.py, network.py, workload.py`

Simulates arbitrary network splits (mountain isolation, multi-island chaos), offline field actions per round, link healing, and transitive convergence.

Stage 5 — 1,000-Trial Proof Harness `simulator/proof_harness.py, demo_stage5.py`

Automated scorecard testing randomized partitions & workloads against 4 invariants (identical state, zero lost updates, max-1 boat alloc, escrow sum conservation).

Stage 6 — Field Security & Forensics `crypto/ciphers.py, signatures.py, keyring.py`

Encrypts sensitive PII fields at rest with AES-256-GCM (PBKDF2-HMAC-SHA256, 100k iterations) and signs events with Ed25519; forged peers are rejected.

Stage 7 — Real TCP Mesh & Laptop CLI `transport/tcp.py, ble_stub.py, run_node.py`

Length-prefixed TCP socket transport over local Wi-Fi/hotspots without internet, standalone interactive CLI runner for 3 laptops, plus BLE transport stub.

Stage 8 — Interactive Command UI `ui/app.py, ui/controller.py, README.md`

Streamlit 3-station Command Center with live SVG mesh map, per-link switches, one-click disaster presets, Ed25519/AES-GCM event inspector, and USB sneakernet JSON sync.

RescueNet — Guarantees, Limitations, Verification & Quick Start

3. Core Guarantees vs. Honest Limitations

Guaranteed by Design

- Strong Eventual Convergence: Identical state after sync.
- Zero Lost Updates: Competing edits stay in conflict ledger.
- Single-Owner Token: Only token holder confirms boat alloc.
- Escrow Conservation: Sum(quotas) + consumed = total.
- Minimal Sync Bandwidth: Version-vector delta exchange.
- Tamper & Theft Defense: Ed25519 + AES-256-GCM PII.

Honest Limitations (CAP Trade-offs)

- Lost Token Device: Destroyed holder locks asset until override.
- Stranded Escrow Quota: Offline node quota stays locked.
- Clock Drift: Bad clocks affect sort order (still converges).
- Pre-Shared Keys: No CA; public keys must be registered.
- Unbounded Log: SQLite grows without snapshot compaction.
- Multi-Laptop CLI: Omit --passphrase unless keys shared.

4. Verification & Test Results

Full Automated Test Suite (python -m pytest tests -v): 39 / 39 PASSED (3.30s)

Stage 1 (test_stage1.py)	**7/7 PASSED**	Deterministic IDs, idempotency, anti-equivocation, order independence
Stage 2 (test_stage2.py)	**4/4 PASSED**	Bi-directional sync, zero-delta overhead, partitions, transitive gossip
Stage 3 (test_stage3.py)	**6/6 PASSED**	Escrow conservation, over-alloc block, quota transfer, boat token, triage
Stage 4 (test_stage4.py)	**5/5 PASSED**	Graph partitions, mountain/valley sync, 5-node chaos, judge scenario
Stage 5 (test_stage5.py)	**5/5 PASSED**	Proof harness, rogue node over-alloc rejection, partition cut integrity
Stage 6 (test_stage6.py)	**4/4 PASSED**	AES-256-GCM PII roundtrip, Ed25519 tamper rejection, raw DB scan
Stage 7 (test_stage7.py)	**4/4 PASSED**	TCP framing, 3-node local TCP convergence, offline timeout, BLE stub
Stage 8 (test_stage8.py)	**4/4 PASSED**	UI compilation, cut/heal/sync workflow, crypto timeline & SVG, USB sync

Additional Checks: - Streamlit server startup health probe returned 200 ok.
- 1,000-Scenario Proof Harness (demo_stage5.py): 1000/1000 scenarios passed across all 4 invariants.

5. Quick-Start Run Instructions

```
# 1. Interactive 3-Node Website (Command Center UI)
cd C:\Users\gowda\.gemini\antigravity\scratch\rescuenet
streamlit run ui/app.py

# 2. Full Automated Test Suite
python -m pytest tests -v

# 3. Three Real Laptops on Offline Hotspot / Wi-Fi (Use each laptop's ipconfig IPv4)
python run_node.py --id alpha --port 8001 --peers beta=192.168.1.11:8002,gamma=192.168.1.12:8003
--db alpha.db
python run_node.py --id beta  --port 8002 --peers
alpha=192.168.1.10:8001,gamma=192.168.1.12:8003 --db beta.db
python run_node.py --id gamma --port 8003 --peers alpha=192.168.1.10:8001,beta=192.168.1.11:8002
--db gamma.db

# Distant Camps (Out of Wi-Fi Range): Use a traveling data-mule laptop, long-range Wi-Fi bridge,
# or the UI's USB Sneakernet JSON Bundle Export / Import tab.
```